

Anurag Sarkar

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EDUCATION

National Institute of Technology, Jamshedpur

Bachelors of Technology in Electronics and Communication Engineering

KVPY Jr. College

Class-12th HSC

December 2021 – April 2025

CGPA : 8.1

April 2019 – April 2021

Percentage : 97.6

EXPERIENCE

MSCI

Technology and Data Intern

January 2025 – June 2025

- Built a scalable log-monitoring system that ingests 2 million+ raw logs from Splunk, classifies them using intrinsic fields with regex fallbacks, groups similar logs via fuzzy matching for cross-environment and historical analysis, and delivers formatted HTML reports through an automated email distribution; optimized high-volume I/O via multithreading with enforced concurrency limits, and used a layered architecture with a State design pattern for flexible templating

MSCI

Technology Analyst

June 2025 – Ongoing

- Designed an agentic RCA system to compensate for incomplete log context by indexing a ~50 GB Python codebase and constructing stack-trace graphs from log origins using regex and intrinsic fields; enabled an AI agent (OpenAI Agents SDK) to traverse controlled depths of method-call graphs, fetch source code at runtime, and query Splunk for additional context to identify root-cause methods and variables.
- Built an AI-driven E2E test-scope generator that integrates with Atlassian (Jira/Confluence) via MCP and Azure DevOps APIs to automatically retrieve issue context, related documentation, and code diffs, enriches it with domain knowledge from a user-provided knowledge base, and generates feature-level test scopes; exposed ADO API to agent through a Python script to control access and embed custom code-retrieval logic.
- Developed an AI agent that generates automated test scripts in a custom Python-based framework, supporting visual, verbose, and tabular prompt formats, and designed the framework to be AI-friendly for reliable code generation and extensibility. Cut down test development time by 91.6%

PROJECTS

Reinforcer-based Generative Adversarial Network for single-image dehazing

- A novel GAN objective function capable of fast and stable training which is facilitated by the use of a reinforcer instead of a discriminator.
- The RbGAN was designed with the sole objective of achieving a framework that allows models to train well on real-world haze datasets which are limited in size and availability.
- The RbGAN performs comparatively well on multiple benchmark datasets (RTTS, NTIRE2018 Image Dehazing Challenge, NH-HAZE, FRIDA).

American Sign Language gesture recognizer with text-to-speech supported user interface | Source Code

- Implementation of **Inception-ResnetV2** which is trained to recognize ASL alpha-numeric gestures.
- The use of inception modules allows the model to perform heavy computations efficiently without facing the Representational Bottleneck problem.
- The Inception-ResnetV2 model is trained with a validation accuracy of **98.62%** and training accuracy of **99.95%**.
- Complete with an interface to upload images into. The output will be relayed via text as well as speech

Garbage Detection model implemented on Raspberry Pi 4 | Source Code

- **MobileNetV2** trained via transfer learning for garbage detection.
- MobileNetV2 was employed due to its novel use of depth-wise convolutions that reduce operational complexity.
- The Raspberry Pi 4 implementation secured 4th place at the **National Innovation Challenge 2.0**.

SKILLS

Languages/Frameworks/Tools: C++, Python, Tensorflow, Docker

Technical: Data Structures and Algorithms, Machine Learning, Deep Learning, Computer Vision, Generative Technology, Edge Machine Learning, Mathematical analysis

EXTRA-CURRICULAR ACTIVITIES

Team Phoenix: Avionics Head

National Service Scheme NIT Jamshedpur: Content Head

Programming Club of NIT Jamshedpur: Member of Machine Learning team